EMERGENCY VEHICLE DETECTION AND TRIGGERING SIGNAL IN SUMO SIMULATOR

Group 25-26J-473

Project Proposal Report

Sangeerthan S -IT22346568

B.Sc. (Hons) in Information Technology Specializing in Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology
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Declaration page of the candidates & supervisor

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

ID Number	Student Name	Signature
IT22346568	Sangeerthan S	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

………………………….

Signature of the supervisor: Date……………...

(Prof Nuwan Kodagoda)

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Signature of the Co-supervisor: Date…………...

(Ms. Karthiga Rajendran)

Abstract

One of the major issues facing emergency services in modern cities is traffic congestion. When moving through heavy traffic, ambulances, fire trucks, and police cars frequently encounter considerable delays, particularly at busy junctions with traditional, fixed-time traffic lights.

As a solution to this problem, the research proposed herein presents the creation of a sophisticated real-time emergency vehicle detection and traffic signal pre-emption system to work in the context of a simulated smart city setup. The system is to be incorporated into the Simulation of Urban Mobility (SUMO) traffic simulator to initiate and provide a visual output.

Its key innovation is the multi-modal detection framework, which utilizes three complementary technologies for accuracy and reliability. YOLO (You Only Look Once), a state-of-the-art deep learning object detection algorithm, will be used for visual detection of emergency vehicles from their physical appearance and structural attributes. Concurrently, audio-based detection methods through machine learning and audio detection will detect the distinctive acoustic signature of siren sounds, allowing detection even when visual information is blocked by other vehicles or environmental factors. HSV (Hue, Saturation, Value) colour space segmentation will also be utilized to detect flashing emergency light bars, which are a highly distinctive and visible attribute of such vehicles, even in low-light or night-time conditions.

Combining these detecting modalities allows the system to recognize emergency vehicles reliably and in real time in a variety of environmental situations, including dim lighting, intense rain, and visual obstructions. After detection, the system uses TraCI (Traffic Control Interface) to interact with the SUMO simulation environment and dynamically modify traffic light conditions, establishing a green corridor for the emergency vehicle that is approaching. The communication between the agents passes the detection details along with the indicator detection details to learn and perform the stimulation activities using the Multi-Reinforcement Learning for the road selection and commanding.

Keywords - YOLO (You Only Look Once), Deep learning, Audio-based detection, Siren sound recognition, Indicator detection, HSV colour space segmentation, Flashing light bar detection

List Of Abbreviations

SUMO	Simulation of Urban Mobility
YOLO	You Only Look Once
HSV	Hue, Saturation, Value
TraCI	Traffic Control Interface
MARL	Multi-Agent Reinforcement Learning
ML	Machine Learning
NN	Neural Network
MFCC	Mel-Frequency Cepstral Coefficients

Table 1 : List of Abbreviations

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1. Introduction

In contemporary cities, urban traffic congestion is a major problem that causes delays, fuel waste, and elevated pollution. Longer travel times and inefficiencies occur from traditional traffic signal systems' set timings, which are unable to adjust to changing traffic conditions. The project's solution to this problem is Smart Traffic Flow Optimization using Multi-Agent Reinforcement Learning (MARL), in which every intersection is represented as an intelligent agent with the ability to dynamically learn and optimize traffic signal patterns. To increase coordination, lessen traffic, and improve road safety, the system uses SUMO simulations to integrate a number of features, such as adaptive pedestrian safety measures, emergency vehicle recognition, and real-time dashboards.

1.1. Background and Literature survey

Emergency vehicle detection is a vital component in modern intelligent transportation systems, particularly within the context of smart cities. The increasing volume of urban traffic often leads to significant delays for ambulances, fire trucks, and police vehicles, especially at busy intersections with fixed-time traffic signals. These delays can have life-threatening consequences, as emergency response time is a critical factor in saving lives and property. [1]

This research explores advanced methodologies for real-time identification and prioritization of emergency vehicles to mitigate traffic congestion and enhance response efficacy. The optimization of emergency vehicle routing and signal pre-emption is paramount for reducing critical response times, particularly within densely populated urban environments where traffic bottlenecks are prevalent. [2] Existing solutions often rely on stationary infrastructure like loop detectors or cameras, which are costly to install and maintain, and offer limited coverage [3].

This research focuses on the development of a real-time, multi-modal emergency vehicle detection system capable of operating in complex and dynamic traffic environments. By integrating the system into the Simulation of Urban Mobility (SUMO) platform, it is possible to simulate and evaluate its ability to trigger traffic signal pre-emption, ensuring a continuous green way for emergency vehicles. The system employs advanced computer vision, audio signal analysis, and colour-based segmentation to detect emergency vehicles under diverse environmental conditions, thereby improving the efficiency and reliability of emergency response operations.

Advantages of the Proposed System

The proposed multi-modal emergency vehicle detection system offers several key expected advantages over traditional detection methods:

- High Accuracy Through Multi-Modal Fusion -Combines YOLO-based visual detection, DSP-based siren sound recognition, and HSV-based light bar segmentation to minimize false positives and false negatives. [4]
- Real-Time Processing -Leverages efficient neural network architectures and optimized audio/visual processing to achieve low-latency detection suitable for live traffic management.
- Improved Emergency Response Times -Enables dynamic green corridor creation, reducing delays for emergency vehicles and potentially saving lives.

1.2. Research Gap

Traditional method of emergency vehicle detection and prioritize them was human based, it requires human energy and cooperative support of other drivers. Higher chance of accidents when crossing red lights or navigating traffic. In the other hand, Upcoming Vehicle Detection (VD) systems rely on vehicle-mounted sensors for Vehicle-to-Vehicle (V2V) communication, requiring every vehicle to be equipped with compatible hardware. These systems operate independently from traffic signals, offering no direct means to reduce waiting times. To satisfy both parties with equal requirements smart based vehicle detection and prioritizing in an interaction way using MARL is the main the planned proposal. The proposed Emergency Vehicle Detection component overcomes these limitations by using multi-modal models' visual recognition, siren sound analysis, and HSV colour segmentation to identify emergency vehicles in real time. Integrated with SUMO for simulation and a dashboard for visualization, it enables traffic signal pre-emption, showcasing dynamic green corridors to authorities for improved emergency response.

Below is a concise gap analysis comparing the system (supervised image classification + HSV light detection + siren audio detection) with existing methods. The analysis highlights limitations of current approaches and novelty contributions of your integrated solution, with comparison of similar research topics with their relevant references.

Table 2 shows the comparison of the technical components that were used in some research papers, every individual component uses to identify specify details using ML models. However, these components encounter failures at certain stages. To overcome these limitations and achieve improved performance, the proposed system adopts a multi-component, microservices-based architecture.

Method	Strengths	Research Gaps	References
Pure Image Classification	High accuracy in controlled environments; real-time inference with lightweight CNNs	❌ Fails in low visibility (fog/night). ❌ Struggles with obscured emergency ❌ Limited to visual range only	[5] [6]
HSV Based Light Detection	Robust for colour-specific detection (red/blue lights); low computational cos	❌ High false positives (e.g., brake lights, traffic signals). ❌ Sensitive to environmental lighting. ❌ Ignores non-visual cues (sirens).	[7] [8]
Siren Sound Detection	Effective for non-line-of-sight scenarios; leverages acoustic uniqueness.	❌ Vulnerable to urban noise pollution (horns, engines). ❌ Limited range. ❌ Cannot classify vehicle type visually.	[9] [10]

Sound + Light Detection	Reduces false negatives via multi-modal data.	❌ Often excludes explicit light-feature extraction. ❌ Underutilizes temporal flashing patterns of lights. ❌ Rarely optimized for emergency vehicles specifically.	[11] [12] [13]

Table 2 : Competitive Analysis

1.3. Research Problem

In modern urban environments, emergency response effectiveness is heavily influenced by the ability of traffic management systems to detect emergency vehicles in real-time and grant them priority passage. Traditional traffic signal systems operate on fixed timers or basic sensor inputs and lack the capability to dynamically adapt for emergency situations. While some systems incorporate visual detection, they often fail under poor visibility, occlusion, or adverse weather conditions. Similarly, audio-based siren detection can be unreliable in noisy urban settings or when the signal-to-noise ratio is low.

Existing approaches tend to rely on a single detection modality either visual, audio, or light-based recognition which limits robustness and accuracy. Moreover, these detection systems are often tested in isolation and not integrated into realistic urban traffic simulations to evaluate their impact on reducing response times. The absence of multi-modal fusion combining computer vision, acoustic recognition, and hue-based light bar segmentation further reduces the adaptability of current solutions.

Additionally, current research rarely integrates real-time detection with a traffic control simulation framework like SUMO to dynamically trigger and optimize signal phases for emergency vehicle clearance. This gap prevents stakeholders from accurately assessing the operational benefits, scalability, and constraints of such systems in complex traffic networks. [13] [4] [14] [15]

2. Objectives

2.1. Main Objective

To design and develop a multi-modal emergency vehicle detection system integrated within the SUMO traffic simulator that uses visual (YOLO + OpenCV), acoustic (siren detection), and hue-based light bar recognition to accurately detect approaching emergency vehicles in real-time and dynamically trigger traffic signal pre-emption to create a clear passage, thereby reducing emergency response times in urban traffic environments.

2.2. Specific Objectives

- Implement Visual Detection -Develop a YOLO-based real-time detection module using OpenCV to identify emergency vehicles from traffic camera feeds under varying lighting and weather conditions.
- Integrate Acoustic Detection -Build and train an ambulance siren recognition module and machine learning techniques to detect emergency sirens in noisy urban environments.
- Apply Hue-Based Light Bar Detection -Implement an HSV colour segmentation and Haar Cascade-based method to detect emergency vehicle light bars and improve detection reliability.
- Integrate with SUMO and TraCI -Connect the detection system to the SUMO traffic simulation via TraCI to dynamically modify traffic signal phases in real-time for emergency vehicle prioritization.
- Evaluate System Performance -Assess detection accuracy, latency, and overall impact on emergency vehicle travel time through simulation experiments under various traffic scenarios.
- Indicator detection -Use to find out the vehicle routes so transfer those details to the neighbour agent to get update on the traffic information.
- Dashboard Visualization - Update the end user with the captured details and data with a promising dashboard and transfer the details to the relevant authority.

3. Methodology

The proposed methodology focuses on developing a multi-modal emergency vehicle detection and traffic signal pre-emption system, integrating visual, acoustic, and hue-based detection modules with the SUMO simulation environment through the TraCI API. The approach is divided into key phases as follows:

3.1. Requirement Gathering

Requirement gathering for this component involves collecting, organizing, and preparing all necessary datasets, system specifications, and functional needs to ensure accurate emergency vehicle detection and effective traffic signal pre-emption within the SUMO simulation environment.

3.1.1. Dataset Collection

To achieve robust detection under varied conditions, a multi-source dataset acquisition strategy will be used:

1. Google Images -Collect a diverse set of emergency vehicle images (ambulances, police cars, fire trucks) from public web sources under varying lighting, weather, and traffic conditions. Images will be filtered for quality, relevance, and resolution before preprocessing.

2. Kaggle Datasets -Utilize publicly available labelled datasets containing emergency vehicles, sirens, and traffic scenes, such as:
    - Emergency Vehicle Image Dataset -High-resolution images with bounding box annotations.
    - Urban Traffic Object Detection Dataset -Includes emergency and non-emergency vehicles for balanced classification.
    - Vehicle Indicator Light Dataset -For turn signal and route prediction modules.
    - Urban Sound Dataset (e.g., UrbanSound8K) -To extract siren audio clips for acoustic detection training.

3. Real-World Image & Audio Capturing –

    Deploy traffic cameras or mobile cameras near intersections to capture actual emergency vehicle footage for fine-tuning detection models.

    Use directional microphones to capture emergency siren audio in different noise conditions, ensuring realistic acoustic training data.

3.2. Development Methodology

3.2.1. Visual Detection Module

Description: Implement a YOLO-based real-time detection module integrated with OpenCV to identify emergency vehicles from live or simulated traffic camera feeds. The model will be fine-tuned with a dataset containing diverse lighting, weather, and traffic conditions.

Technologies & Tools:

- YOLOv8 / YOLOv5 (PyTorch) -Object detection framework for real-time identification.
- OpenCV -For video frame extraction, preprocessing, and drawing detection bounding boxes.
- Python -Core programming language for detection and integration.
- CUDA -For GPU acceleration during detection tasks.

3.2.2. Audio Detection Module

Description: Develop a siren detection system using Audio detection and machine learning to recognize ambulance sirens in noisy urban soundscapes. Audio input will be processed to extract features such as MFCC for model training.

Technologies & Tools:

- Librosa -For audio signal processing and feature extraction.
- TensorFlow / Keras -For building and training the siren classification model (CNN/LSTM).
- Scikit-learn -For preprocessing and baseline ML models.

- PyAudio -For capturing audio streams in real-time.

3.2.3. Hue-Based Light Bar Detection Module

Description: Implement HSV colour segmentation to detect characteristic emergency vehicle light bars, combined with Haar Cascade classifiers for improved detection reliability under varying conditions. Figure 1 shows the sound detection of siren using NN to detect it and give the single output.

Technologies & Tools:

- OpenCV -For HSV colour space segmentation and image filtering.
- Haar Cascade Classifiers -Pre-trained or custom-trained for light bar detection.
- Python -For scripting and integration with other modules.

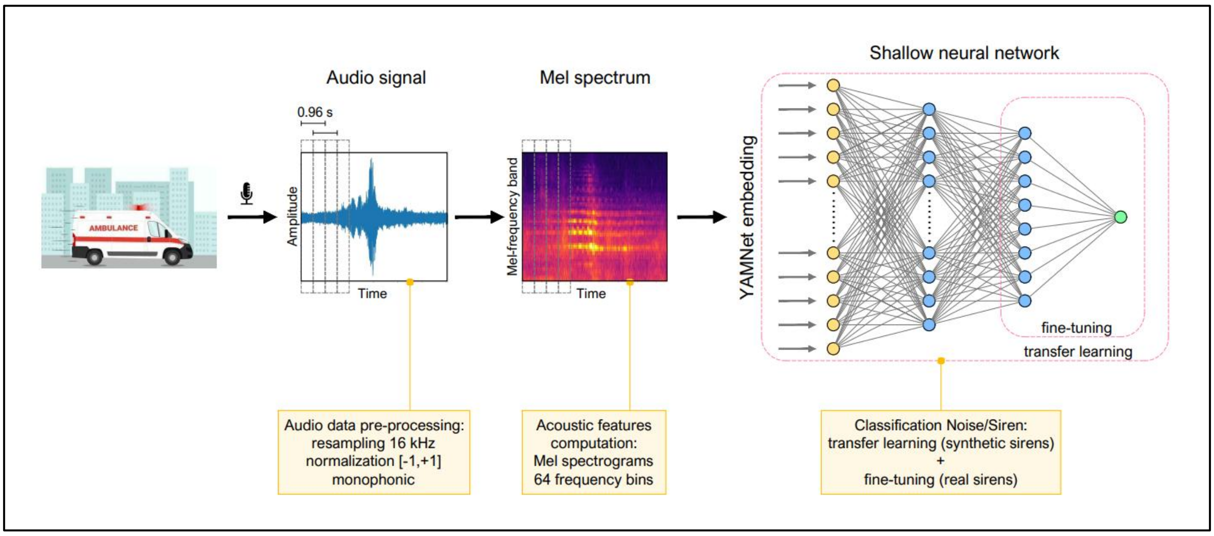

Figure 1 : Audio Siren Detection Source [14]

3.2.4. SUMO and TraCI Integration

Description: Connect the detection system to the SUMO traffic simulation environment using the TraCI API to trigger and adapt traffic light phases in real-time upon detecting an approaching emergency vehicle.

Technologies & Tools:

- SUMO (Simulation of Urban Mobility) -Traffic simulation software.

- TraCI (Traffic Control Interface) -Python API for controlling simulation events.
- NetEdit -SUMO network editor for creating and configuring intersections.

Figure 2 is the virtual environment where the traffic-based programs can be done it is more flexible and portable to use.

Figure 2 : SUMO Stimulator , Source : https://networksimulationtools.com/sumo-simulator/

3.2.5. Indicator Detection & Route Prediction

Description: Detect vehicle turn indicators to predict movement patterns and intended routes, enabling proactive traffic flow adjustments and informing neighbouring intersections or control agents.

Technologies & Tools:

- OpenCV -For indicator light detection and flashing pattern recognition.
- YOLO / Haar Cascade -For vehicle frontal/rear view detection prior to indicator analysis.

3.2.6. Dashboard Visualization & Data Sharing

Description: Develop a dashboard to display detected emergency vehicle data, system alerts, traffic light status changes, and performance metrics in real-time. Data will be transmitted to relevant authorities for action.

Technologies & Tools:

- Flask / FastAPI -Backend API to serve detection and simulation data.
- React.js -Frontend dashboard for real-time visualization.

3.2.7. System Performance Evaluation

Description: Evaluate the integrated system's performance within SUMO simulations under varying traffic densities, weather conditions, and noise levels. Metrics will include detection accuracy, false positive/negative rates, system latency, and impact on emergency vehicle travel time.

Technologies & Tools:

- Python (NumPy, Pandas) -For result logging and statistical analysis.
- Matplotlib / Seaborn -For visual representation of evaluation metrics.

These steps will be tracked with the proper project management tool. The development of the Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component will follow the Agile software development methodology, enabling iterative, flexible, and feedback-driven progress. Agile was selected due to the project's complexity, multi-modal nature, and the need to integrate diverse technologies (YOLO, DSP-based siren detection, HSV segmentation, SUMO simulation) while ensuring adaptability to changes.

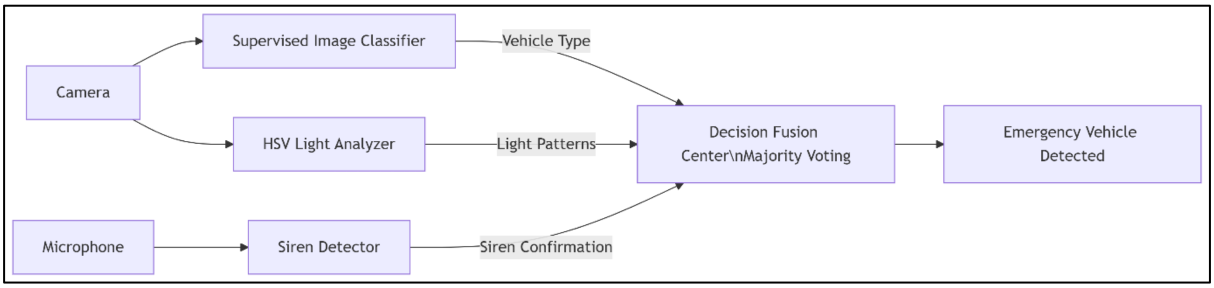

Figure 3 :Component Flow Diagram

3.3. Project Management Methodology

Research requires effective project management, where scope, time, and cost must be carefully aligned through a structured system or procedure. For the adaptive quiz platform project, a process-oriented methodology will be employed, dividing the project into distinct stages with meticulously planned inputs and outputs. This approach is designed to accommodate changes, identify risks, and manage the project effectively. JIRA will be used to support this methodology, providing a robust platform for tracking progress, managing tasks, and ensuring that project requirements and timelines are met efficiently.

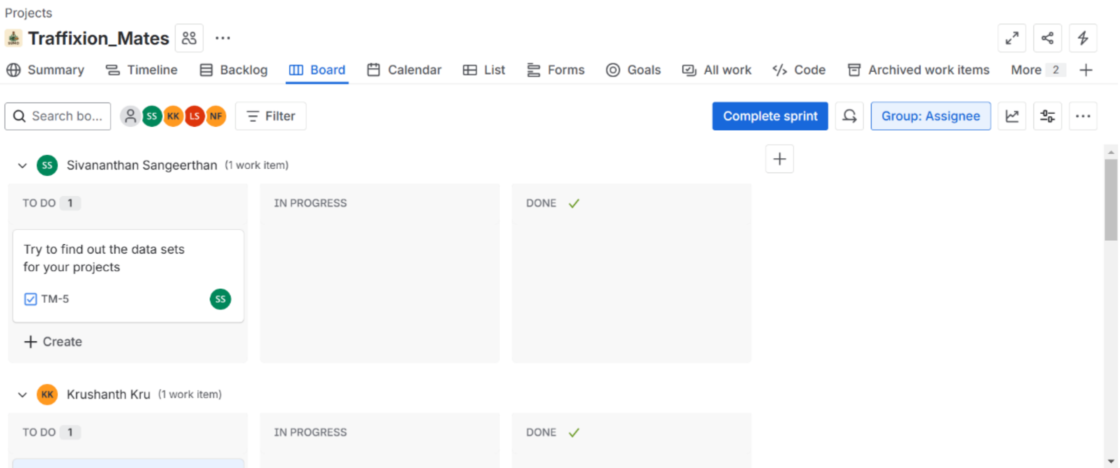

Figure 4 : Jira management tool

3.4. Feasibility Study

The feasibility study assesses the practicality of implementing the proposed Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component in terms of technical capabilities, project scheduling, and economic viability.

3.4.1. Technical Feasibility

The proposed system integrates computer vision, audio signal processing, colour segmentation, and traffic simulation technologies. The following technical resources and competencies are available within the team to ensure successful development:

Knowledge on Technologies

To develop the proposed solution, the knowledge in below mentioned technologies should be wanted.

- Computer Vision: Proficiency in YOLOv8/YOLOv5 and OpenCV for object detection and image processing.
- Acoustic Signal Processing: Experience with Librosa , and machine learning models for siren recognition.
- Colour-Based Detection: Knowledge of HSV segmentation and Haar Cascade classifiers for light bar recognition.
- Traffic Simulation: Hands-on experience with SUMO and TraCI API for real-time signal control.
- Machine Learning: Model training and fine-tuning using PyTorch and TensorFlow.

Knowledge on Tools

To develop the proposed optimization model, all members should have quite an in-depth understanding of the development tools and project management, as well as other supportive tools. We will be using Jira to facilitate project tracking and management throughout the development process.

- Development Tools: VS Code, Jupyter Notebook, Google Colab (GPU-enabled training).
- Project Management: Jira for sprint tracking and collaboration.
- Annotation Tools: LabelImg and Roboflow for dataset labelling.
- Dashboard Tools: Flask/FastAPI , React.js , Chart.js/D3.js .

Data Collection & Preprocessing Skills

- Ability to collect datasets from Google Images, Kaggle, and real-world recordings.
- Experience in image augmentation, audio denoising, and annotation to improve model robustness.

3.4.2. Schedule Feasibility

The project will follow the Agile methodology, divided into 2–3 week sprints for each module:

- Dataset preparation (Sprint 1)
- Visual detection (Sprint 2)
- Acoustic detection (Sprint 3)
- Hue-based detection (Sprint 4)
- SUMO & TraCI integration (Sprint 5)
- Indicator detection (Sprint 6)
- Dashboard development (Sprint 7)
- Testing & optimization (Sprint 8)

Given the modular approach, the timeline is realistic and allows flexibility for refinement based on test feedback. Milestones will be tracked in Jira, ensuring tasks are delivered within deadlines.

3.5. System Analysis

3.5.1. Software Solution Approach

The proposed system will be developed as a modular, multi-modal detection and simulation platform that integrates visual, acoustic, and hue-based detection techniques with the SUMO traffic simulation to enable real-time emergency vehicle prioritization.

Data Collection -Gather emergency vehicle image datasets (Google, Kaggle, real-world) and siren audio datasets (UrbanSound8K, Kaggle, recordings).

Pre-processing -Augment and normalize images.

Model Training -Use YOLOv8/YOLOv5 for visual detection, HSV + Haar Cascade for light bar detection.

Multi-Modal Fusion -Combine visual, audio, and hue-based outputs via decision fusion to improve accuracy.

SUMO Integration -Control traffic signals dynamically with TraCI when emergency vehicles are detected.

Route Prediction -Detect turn signals to predict paths and share with nearby intersections.

Dashboard -Real-time monitoring via React.js frontend and Flask/FastAPI backend, with charts/maps.

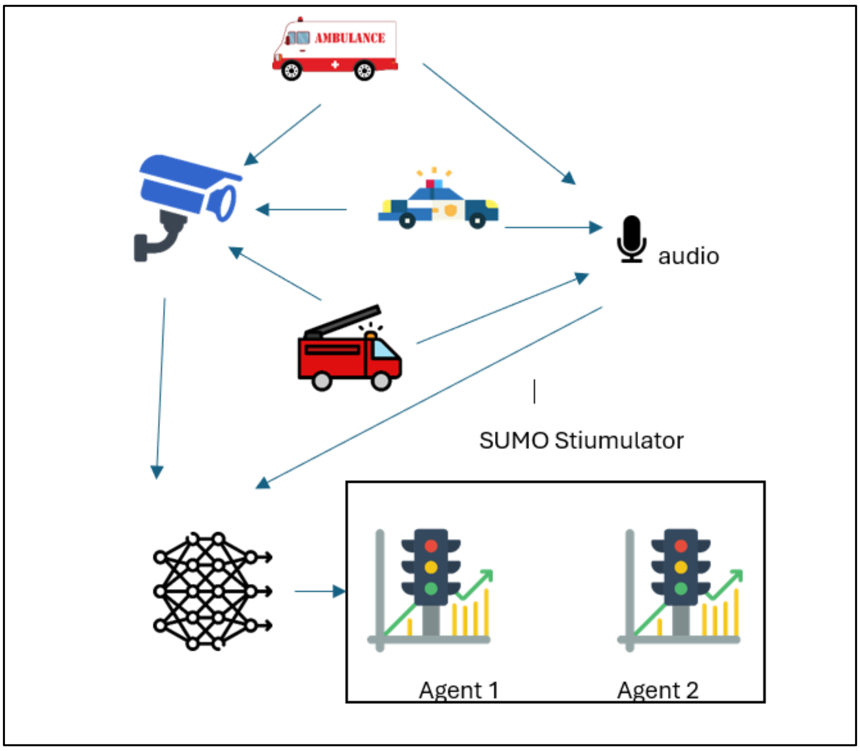

Figure 5 : System Diagram

3.5.2. Tools and Technologies

The proposed system will be developed using a combination of software frameworks, programming languages, simulation tools, and development platforms to ensure robustness, scalability, and real-time performance.

Programming Languages

- Python 3.x -The core development language for computer vision, audio processing, machine learning, and SUMO integration due to its extensive library ecosystem.
- JavaScript (React.js) -For building the real-time interactive dashboard frontend

Frameworks and Libraries

- YOLOv8 / YOLOv5 (PyTorch) -State-of-the-art deep learning frameworks for real-time visual emergency vehicle detection.
- OpenCV -Computer vision library for video frame processing, HSV colour segmentation, and Haar Cascade implementation.

- TensorFlow / Keras -Machine learning frameworks for training CNN/LSTM models for siren sound recognition.
- Librosa -Python library for audio feature extraction such as MFCC, used in acoustic detection.
- Scikit-learn -For data preprocessing, model evaluation, and integration of classical ML methods.

Traffic Simulation and Integration Tools

- SUMO (Simulation of Urban Mobility) -Microscopic, open-source traffic simulator for testing signal pre-emption strategies in urban environments.
- TraCI API -Traffic Control Interface for real-time control of SUMO simulations through Python scripts.
- NetEdit -SUMO's network editor for building and modifying traffic network layouts.

Data Collection and Annotation Tools

- Google Images / Kaggle Datasets -Sources for diverse emergency vehicle, siren, and traffic datasets.
- LabelImg / Roboflow -Annotation tools for creating bounding box labels for training YOLO models

Backend Development and API Tools

- Flask / FastAPI -Python-based web frameworks for creating RESTful APIs to serve detection and simulation data to the dashboard.

Frontend Development and Visualization Tools

- React.js -For creating a responsive, real-time web dashboard for visualizing detections and simulation status.

Development and Project Management Tools

- VS Code -Main integrated development environment for coding and debugging.
- Google Colab -For GPU-enabled training of detection and audio models.
- Jira -For Agile sprint planning, backlog management, and task tracking.
- Git & GitHub -For version control and collaborative development.

Testing Tools

- Pytest -For automated unit and integration testing of Python modules.
- Postman -For testing and validating API endpoints.

Hardware Requirements

- GPU-enabled Workstation / Cloud GPU -For training deep learning models efficiently.
- CCTV / IP Cameras -For real-world visual data collection and live feed integration.
- Directional Microphones -For real-time siren sound recording in urban environments.

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3.6. Project Requirements

The project requirements define the expected functionality and operational qualities of the Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component. These requirements ensure that the system meets both performance goals and real-world applicability.

3.6.1. Functional Requirements

The system shall:

- Visual Detection -Accurately detect emergency vehicles in real-time using a YOLO-based model and OpenCV video processing.
- Acoustic Detection -Identify ambulance sirens using DSP techniques and machine learning models even in noisy urban environments.
- Hue-Based Light Bar Detection -Detect emergency vehicle light bars using HSV colour segmentation and Haar Cascade classifiers.
- Multi-Modal Fusion -Combine visual, acoustic, and hue detection outputs to increase detection accuracy and reduce false positives.
- Indicator Detection -Detect turn indicators of emergency vehicles to predict movement routes.
- SUMO Integration -Communicate with SUMO via TraCI API to pre-emptively adjust traffic light phases for emergency vehicle passage.
- Route Sharing -Transfer predicted routes to neighbouring simulated intersections for coordinated signal adjustments.
- Dashboard Visualization -Display real-time detection events, simulation updates, and traffic light statuses on a web-based dashboard.
- Data Logging -Store detection events, simulation changes, and performance metrics for later analysis.
- Authority Notification -Send alerts to relevant traffic management authorities when emergency vehicles are detected.

3.6.2. Non-Functional Requirements

The system shall:

- Performance -Operate with a detection latency of less than 500 ms per frame for visual and acoustic detection combined.
- Accuracy -Achieve an overall multi-modal detection accuracy of at least 95% in simulation testing.
- Scalability -Support integration into larger Intelligent Transportation Systems (ITS) with minimal modifications.
- Reliability -Maintain consistent detection performance under varying lighting, weather, and noise conditions.
- Security -Protect stored and transmitted data through encryption and secure communication channels.
- Maintainability -Ensure modular code structure with proper documentation for easy updates and debugging.
- Usability -Provide an intuitive dashboard interface that can be operated without advanced technical knowledge.
- Compatibility -Be deployable on both local GPU-enabled systems and cloud-based environments.
- Resource Efficiency -Optimize computational resource usage to run on mid-range hardware without significant performance drops.

3.7. Project Scope

The scope of this project defines the boundaries and extent of the Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component, outlining the features and functionalities that will be developed and those that will not be included in the current phase.

3.7.1. In-Scope

The system will include:

Real-Time Visual Detection -Detect emergency vehicles using YOLO-based object detection integrated with OpenCV.

Acoustic Siren Detection -Identify ambulance sirens through DSP-based feature extraction and ML models.

Hue-Based Light Bar Recognition -Detect light bars using HSV segmentation and Haar Cascade classifiers for improved reliability.

Multi-Modal Fusion -Combine detection results from visual, acoustic, and hue-based modules to enhance accuracy.

Indicator Detection and Route Prediction -Identify turn indicators and predict possible routes of emergency vehicles.

SUMO Integration -Control traffic signals dynamically using TraCI API in the SUMO simulation to prioritize emergency vehicle passage.

Route Sharing Between Intersections -Transfer predicted emergency vehicle route data to neighbouring simulated traffic controllers.

Dashboard Visualization -Provide a real-time, web-based dashboard to display detection events, simulation status, and traffic light changes.

Testing in Simulated Environments -Evaluate system performance under various simulated traffic densities, weather, and noise conditions.

3.7.2. Out-of-Scope

The system will not include:

Real-World Deployment -The current project will focus on SUMO simulation and will not control actual traffic lights in physical intersections.

Integration with V2X Communication Systems -Vehicle-to-everything communication is excluded in this phase but may be considered in future enhancements.

Advanced Gamification or User Training Features -No points, leaderboards, or driver training modules will be implemented.

Offline Functionality -The system will require live video/audio streams or recorded datasets and will not operate in a fully offline mode.

Integration with Live City Traffic Databases -Only simulated data from SUMO will be used for traffic control and testing.

3.8. Testing

Testing is a critical phase in the development of the Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component to ensure system reliability, accuracy, and real-time performance. Multiple testing approaches will be applied to validate individual modules, integrated functionality, and overall simulation outcomes.

3.8.1. Unit Testing

Purpose: To validate that each independent module of the system works as intended in isolation before integration.

Scope:

- Visual Detection Module: Verify YOLO model correctly detects emergency vehicles in test images and videos with expected accuracy.
- Acoustic Detection Module: Ensure siren recognition model correctly classifies audio clips as siren or non-siren.
- Hue-Based Detection Module: Test HSV segmentation and Haar Cascade to ensure accurate light bar detection under different lighting conditions.
- Indicator Detection Module: Confirm detection of turn signals and extraction of route prediction data.

3.8.2. Integration Testing

Purpose: To verify that all modules work together correctly when combined into the complete system.

Scope:

- Integration of visual, acoustic, and hue-based detection outputs into the multi-modal fusion logic.
- Communication between detection modules and the SUMO simulation via TraCI API.
- Route prediction data transfer to neighbouring simulated intersections.
- Real-time update flow from backend to dashboard using REST API.

3.8.3. Performance Testing

Purpose: To assess the system's responsiveness, accuracy, and computational efficiency.

Metrics:

- Detection Latency: Measure the time taken from receiving visual/audio input to generating detection results.
- Detection Accuracy: Evaluate multi-modal fusion accuracy against a ground-truth dataset.
- System Throughput: Test number of frames processed per second (FPS) under GPU and CPU configurations.
- Resource Utilization: Monitor CPU, GPU, and memory usage during peak operation.

4. Work Breakdown

This project develops an emergency vehicle detection system for the SUMO traffic simulator, structured into clear phases. It begins with planning and gathering requirements. The core involves building specialized detection modules using vision, sound, and light recognition, fused by an algorithm to trigger traffic signal pre-emption via SUMO's TraCI. The system undergoes rigorous unit, integration, and performance testing before final deployment in the simulator. The project concludes with comprehensive documentation and evaluation of results against initial objectives.

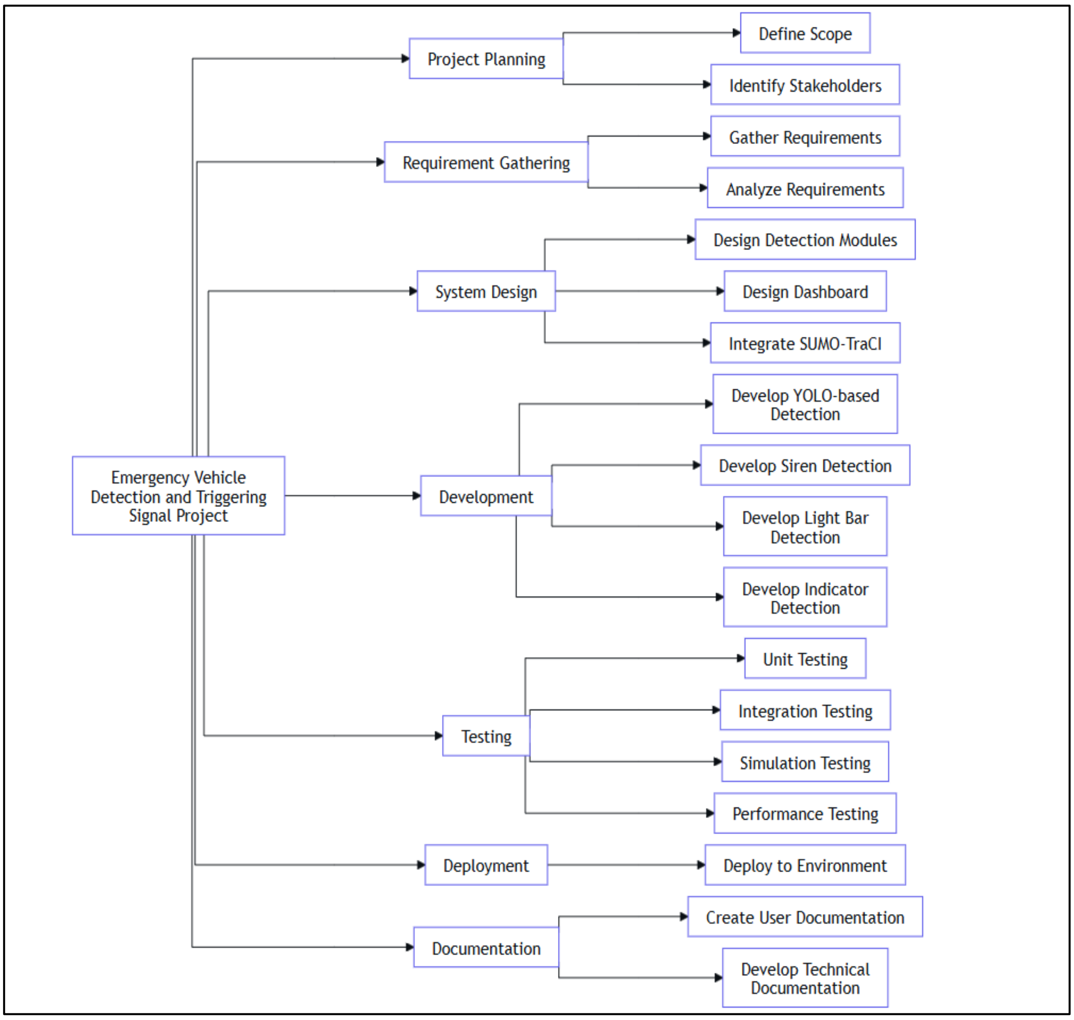

Figure 6 : Work Breakdown Structure

5. Expected Outcomes

The Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component is expected to deliver measurable improvements in emergency response efficiency through accurate detection, fast decision-making, and intelligent traffic control. The outcomes will focus on performance, usability, and impact within simulated urban traffic environments.

Measurable Performance Goals

High Detection Accuracy

- Achieve ≥ 95% accuracy in detecting emergency vehicles using multi-modal fusion.
- Reduce false positives and false negatives to < 5% during simulation tests.

Low Detection Latency

- Process visual and audio input streams with an average latency below 500 ms.

Effective Traffic Signal Pre-emption

- Demonstrate successful coordination between multiple intersections via route-sharing.

Functional Outcomes

- A fully functional YOLO-based real-time visual detection module integrated with OpenCV.
- A ML-based acoustic siren detection module capable of operating in noisy urban environments.
- An HSV segmentation and Haar Cascade light bar detection module for redundancy in visual recognition.

6. Budget and Budget Justification

The cost of development will remain low, as the system primarily uses open-source frameworks and tools. The main expenses will involve data collection equipment and hosting services.

Estimated Budget:

Type	Cost (LKR)	Remarks
GPU-enabled computing (Cloud/Colab Pro)	8,000	Training model fine-tuning
Data Collection (camera, microphone)	20,000	Real-world capturing
Internets and hosting	6,000	Hosting dashboard
Stationery and documentation	1,000	Printing and report costs
Total	35,000	-

Table 3 : Budget and Budget Justification

The low-cost structure, combined with the scalability of the system, makes the project economically viable. The solution can later be commercialized or integrated into existing Intelligent Transportation Systems (ITS) for real-world deployment.

7. Timeline

The project will proceed through a well-organized development timeline. At the outset, the team will concentrate on completing the platform's design, including creating a comprehensive system diagram and choosing the essential technologies. Next, the implementation phase will commence, focusing on fine-tuning and developing algorithms for SUMO Stimulation. Simultaneously, performance analysis tools and other system components will be integrated. The final phase will include extensive testing, performance assessment, and the formulation of commercialization plans. Throughout these stages, the project will emphasize careful development and validation before the platform's official launch.

The components require data and model training with fine tuning process in the case the Figure 8 shows the time allocation for the whole year form the beginning.

	2025						2026				
Task	July	August	September	October	November	December	January	February	March	April	May
Feasibility study	■										
Evaluate feasibility and background study	■										
Requirement gathering	■										
Equipment gathering and testing		■									
Background survey		■									
Literature review		■									
Requirement analysis		■									
Software requirement specification		■									
Functional, non-functional requirement		■									
Proposal presentation		■									
Project proposal report		■									
Data collection		■									
ML model creation			■								
Model training			■								
Model tunning			■	■	■						
Designing wireframes					■	■	■				
SUMO and Software integration					■	■	■				
Deployment & maintenance					■	■	■				
Progress presentation 1						■					
Research paper						■	■	■			
Progress presentation 2								■			
Testing									■	■	
Final presentation and viva											■

Figure 7 : Component Timeline

8. Communication Management Plan

The Communications Management Plan is designed to ensure that all team members, as well as the supervisor and co-supervisor, receive the necessary information to effectively perform their roles throughout the project. Successful project execution relies on strategic planning and effective communication. This plan outlines how to communicate with various stakeholders efficiently, detailing the audience, content, format, frequency, and expected outcomes of communications. It also specifies each stakeholder's role, task assignment, and communication strategy based on their influence, interests, and expectations.

Communication Objectives

Proactive communication is essential for project success. Communication should be:

• Adequate: Providing information in the right format and content.

• Specific: Tailored to the targeted audience.

• Sufficient: Including all necessary details.

• Concise: Clear and to the point, avoiding unnecessary repetition.

• Timely: Addressing relevant points at appropriate times.

Communication Media The following media will be utilized for communication throughout the project:

1.Microsoft Teams : For formal updates, documentation, and detailed communications.

2. Documents (MS Word and PowerPoint): For reports, presentations, and detailed project documentation.

3. Phone Calls: For quick discussions and urgent matters.

4. Meetings: Conducted via meeting rooms, conference phones, or Microsoft Teams for in-depth discussions, planning, and progress reviews.

5. Chats (WhatsApp): For informal, quick exchanges and real-time updates.

9. Commercialization

The Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component has strong potential for commercialization in both public sector and private sector markets. The system addresses a critical urban challenge reducing emergency response times making it valuable for deployment in Intelligent Transportation Systems **(ITS)** worldwide.

Potential Markets

- Government Traffic Management Authorities
  - City councils and transportation departments can integrate the system into existing traffic control centres to optimize emergency vehicle movement.
- Emergency Services
  - Ambulance, fire, and police departments can deploy the solution to ensure faster and safer passage through congested urban intersections.
- Smart City Solution Providers
  - Technology vendors offering IoT-based traffic management systems can incorporate the multi-modal detection module as a premium feature.

Future Commercial Potential

The technology can be extended beyond emergency vehicles to support:

- Public Transport Priority: Buses and trams receiving traffic signal priority during peak hours.

- Autonomous Vehicle Integration: Providing signal priority for driverless emergency and service vehicles.

10.Summary

The proposed Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component is a multi-modal intelligent traffic management solution designed to prioritize emergency vehicles in urban environments. By integrating YOLO-based visual detection and machine learning-based siren recognition, and HSV Cascade light bar detection, the system achieves high detection accuracy under varying conditions. Real-time integration with SUMO and TraCI enables dynamic traffic signal pre-emption, reducing response times and enhancing road safety.

The project follows an Agile development methodology, ensuring iterative improvement through testing phases including unit, integration, simulation, and performance testing. Expected outcomes include reduced travel times for emergency vehicles, improved intersection safety, and a scalable foundation for real-world deployment. With strong commercialization potential in government traffic control, emergency services, and smart city solutions, the system offers a cost-effective, high-impact contribution to modern urban mobility management.

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